

MUSTAFA IQBAL

SOFTWARE ENGINEER

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

EXPERIENCE

Independent Software Engineer | 2024-Present

Project-based product and client work | Remote

- Built software for early-stage teams, working directly with founders and domain experts to turn operational requirements into deployed products and third-party integrations.
- Contributed backend and frontend features to a production AI SaaS on an approximately 20-person team using TypeScript, Node.js, and Prisma, including billing credits, model routing, retrieval, background queues, and object storage.

AI Research Intern & Team Lead | Jun-Aug 2025

Genesys Research Lab | Islamabad, Pakistan

- Led four engineers building TalentFlow, an internal recruitment platform combining five candidate sources, semantic candidate-role matching, and conversational screening.
- Built asynchronous FastAPI services, a Weaviate and Nomic retrieval pipeline, and a Llama 3 interview agent; deployed the system with Docker Compose on lab infrastructure.

PROJECTS

[Revy](#) | Live on SimplaBots

TypeScript, Next.js, PostgreSQL, Google Business Profile API, OpenAI

- Built a Google Business Profile review-response agent that syncs reviews, constructs business and location context, and supports manual drafting, approval before posting, or automatic posting of eligible positive reviews.
- Implemented Google OAuth and token refresh, paginated idempotent ingestion, a Prisma/PostgreSQL data model, pg-boss jobs, and an operator dashboard for locations, drafts, approvals, and publishing.

[Emmy](#) | Live on SimplaBots

TypeScript, Next.js, PostgreSQL, Gmail API, Pub/Sub, Inngest, OpenAI

- Built a Gmail categorization agent that classifies messages using sender and recipient metadata, user-defined categories, contact-group constraints, and prior corrections.
- Implemented Google OAuth, Pub/Sub and Inngest processing, controlled concurrency, native Gmail label writes, thread-level correction handling, and per-email decision logs surfaced in the dashboard.

[MelodyMind](#) | Final-Year Project

Python, PyTorch, FastAPI, Pinecone, PostgreSQL, React Native

- Co-built a multimodal music companion that generates emotion-aware playlists from text, images, and voice, with a conversational voice DJ and asynchronous Demucs stem separation.
- Built the research and application stack as a three-person team: a PyTorch CLAP-to-Nomic projection head trained with InfoNCE, FastAPI services, Pinecone retrieval, PostgreSQL, and a React Native client.

[ArchPHI](#) | Co-founder

Python, ezdxf, Shapely, PyMuPDF, LLM agents

- Building a CAD takeoff product where an agent writes Python measurement programs against searchable drawing data, returning quantities grounded in the underlying CAD and flagging items the package cannot support.

EDUCATION

B.S. Computer Science, FAST-NUCES, Islamabad | 2022-2026

CGPA: 3.38/4.00 | Dean's List: 3 terms

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Bash, C++, Java

AI & Machine Learning: AI agents, tool use, context engineering, prompt engineering, LLM evaluation, RAG, embeddings, vector search, PyTorch, LangChain, OpenAI, Gemini, and Groq APIs

Backend & Data: Node.js, FastAPI, REST APIs, PostgreSQL, Supabase, Prisma, SQLAlchemy, OAuth 2.0, JWT, webhooks, Pinecone, Weaviate, background jobs, Pub/Sub, Inngest, pg-boss

Frontend & Deployment: React, Next.js, React Native, Tailwind CSS, Docker, Vercel, Railway, GitHub Actions, Git